Bow Valley College

# Course Outline

## DATA3409

Data Visualization II
Winter 2023 - Current

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# DATA3409 Data Visualization II

## COURSE DESCRIPTION

In this course, learners apply a range of techniques to realize any design vision. Learners integrate their technical and communication skills to build and refine simple and complex visualizations to optimize effectiveness. Finally, learners close off a project, ensuring that decision makers have what they need to make decisions. Additional Performance Standards: This course, along with Designing Effective Visualizations, make up the Visualizing Data and Insights competency; competency assessments in both courses must be successfully completed to be deemed competent.

| | |
|---|---|
| REQUISITES | None |
| EQUIVALENTS | None |
| CREDITS | 3 |
| HOURS | 45 |
| ELIGIBLE FOR PLAR | No |

## COURSE COMPETENCY

### COMPETENCY TITLE

Data Management and Analytics – Visualizing Data and Insights

### COMPETENCY STATEMENT

Data Management and Analytics professionals convert data insights into powerful, evidence-based visual narratives, complemented by other communication techniques, to ensure stakeholders can efficiently and effectively translate analysis results into business decisions.

### COMPETENCY DESCRIPTION

Data analysts and scientists are required to present insights in a meaningful and compelling way so that stakeholders can fulfill business objectives. In order to facilitate decision making, they must present data, relationships, and recommendations in ways that are intuitive to all stakeholders, regardless of technical background. Data professionals who perform the visualizing data and insights job function incorporate the principles of visual storytelling to design visualization elements, reports, and interfaces (e.g. dashboards) that meet stakeholder needs and support decision making. They apply a range of techniques to realize any design vision, integrating their technical and communication skills to build and refine simple and complex visualizations to support timely and effective decision making. Data visualization bridges the gap between the data professional and the organizational decision maker, making the visualization data and insights job function an in-demand competency for anyone working in a business-facing data analysis role.

The Visualizing and Presenting Insights competency makes up part of the Data Management and Analytics job profile along with:

- Designing Data Projects
- Acquiring and Wrangling Data
- Performing Data Analysis

## COURSE LEARNING OUTCOMES

Bow Valley College is committed to ensuring our graduates can demonstrate their abilities in key areas that will make them effective citizens and encourage their development as lifelong learners. In addition to the discipline-specific skills that learners acquire in their programs, the College has identified ten learning outcomes.

**College-Wide Outcomes:**

1. Communication
2. Thinking Skills
3. Numeracy and Financial Literacy
4. Working with Others
5. Digital Literacy
6. Positive Attitudes and Behaviours
7. Continuous Learning
8. Health and Wellness Awareness
9. Citizenship and Intercultural Competence
10. Environmental Sustainability

| # | COURSE LEARNING OUTCOME(S) | COLLEGE WIDE OUTCOMES SUPPORTED |
|---|---|---|
| 1 | Create a dashboard to present a meaningful and cohesive narrative by demonstrating visual and written communication skills. | 1, 2, 3, 5, 6 |
| 2 | Present insights to stakeholders by demonstrating attention to detail and communication skills. | 1, 2, 3, 5, 6 |

## LEARNING PATHWAY

**The time it takes learners to demonstrate competencies will vary. An example of a suggested schedule for learning and development is shown below. Learners will need to plan out their assessment attempts within their course. For additional information, please consult the Course Offering Information in Brightspace.*

| WEEK/HOURS | LEARNING AND DEVELOPMENT PLAN |
|---|---|
| Week 1 | Review Performance Demonstration, Competency Assessment, and learning pathway |
| Week 2 | Create Success Plan; Complete Competency Assessments |
| Week 3 | Create visualizations |
| Week 4 | Create visualizations |
| Week 5 | Combine visualizations to make a dashboard |
| Week 6 | Combine visualizations to make a dashboard |
| Week 7 | Complete a quality review of the dashboard |
| Week 8 | Publish and share the dashboard |
| Week 9 | Reading Week |
| Week 10 | Create a summary presentation and report |
| Week 11 | Create a summary presentation and report |
| Week 12 | Dashboard |
| Week 13 | Dashboard |
| Week 14 | Presentation and Report |
| Week 15 | Performance Demonstration and Competency Assessment |

## ASSESSMENT

This course follows an assessment-first approach, in which learners will be assessed, and receive structured feedback, and a personalized learning plan. Learners will also receive differentiated support from an instructor based on their individual needs.

Learners will have a variety of ways to demonstrate they have met the required competency through the demonstration of learning outcomes and criteria as laid out in the rubric. Learners will have multiple (but not unlimited) attempts to prove competency. It is suggested that learners plan out their assessment attempts within their course.

Learners will have flexibility in how they satisfy course learning outcomes while still adhering to the criteria found in the rubric and the Course Offering information. Please refer to the Course Offering Information and the rubric in Brightspace for additional information.

| COURSE LEARNING OUTCOMES | ASSESSMENT |
|---|---|
| 1, 2 | Performance Demonstration |

## PERFORMANCE STANDARDS

Successful demonstration of all learning outcomes is required to pass this course. This course may be part of a competency or larger program of study.

This course, along with DATA3408: *Data Visualization I,* make up the Visualizing Data and Insights competency. Successful completion of the performance demonstration(s) in both courses and competency assessment is necessary to be deemed competent in Visualizing Data and Insights competency. In order to be successful in this course, learners must be competent in the Math for Data Analytics and Data Programming course outcomes.

Please consult with the program area or the Bow Valley College website for completion requirements.

## GRADING SCHEME

| Symbol | Description | Value in GPA |
|---|---|---|
| P | Competent (Pass) – The Learner has demonstrated the required elements and has met the course competencies. | N/A |
| F | Developing (Fail) - The learner has not yet demonstrated the required elements and has not met the course competencies. | N/A |

## REQUIRED LEARNING RESOURCES

Additional learning resources may be found in the Course Offering Information or in Brightspace.

## ADDITIONAL INFORMATION

Knaflic, C. (2019). *Storytelling with data: Let's practice*. Wiley.
ISBN: 9781119621492 print copy

ISBN: ISBN: 9781119621508 ebook

Additional information may be found in the Course Offering Information or in Brightspace.

## ACADEMIC ACCOMMODATIONS

Learners with a disability (learning, physical, and/or mental health) may qualify for academic and exam accommodations. For more information, or to apply for accommodations, learners should make an appointment with Accessibility Services in the Learner Success Services (LSS) Department. Accessibility Services can also assist learners who may be struggling with learning but do not have a formal diagnosis. To make an appointment visit LSS on the first floor of the south campus or call 403-410-1440. It is the learner's responsibility to contact Accessibility Services and request academic accommodations. For more information, please visit our website at http://www.bowvalleycollege.ca/accessibility.

## INSTITUTIONAL POLICIES

Bow Valley College is committed to the highest standards of academic integrity and honesty. Learners are urged to become familiar with and uphold the following policies: Academic Integrity (500-1-7), Learner Code of Conduct, Procedures and Guidelines (500-1-1), Learner Appeals (500-1-12), Attendance (500-1-10), Grading (500-1-6), Academic Continuance and Graduation (500-1-5), and Electronic Communications (300-2-13). Audio or video recording of lectures, labs, seminars, or any other teaching and learning environment by learners is allowed only with consent of the instructor as part of an approved accommodation plan. Recorded material is to be used solely for personal study and is not being used or distributed without prior written consent from the instructor.

**Turnitin:**

Students may be required to submit their course work to Turnitin, a third-party service provider engaged by BVC. Turnitin identifies plagiarism by checking databases of electronic books and articles, archived webpages, and previously submitted student papers. Students acknowledge that any course work or essays submitted to Turnitin will be included as source documents in the Turnitin.com reference database, where it will be used solely to detect plagiarism. The terms that apply to a student's use of Turnitin are described on Turnitin.com.

**Online Exam Proctoring:**

Examinations for this course may require proctoring through an online proctoring service. Online proctoring enables online exam taking within a controlled and monitored environment, thereby enhancing academic integrity. Online proctoring may occur through a variety of methods, including but not limited to:

a. live online proctoring where a remote invigilator authenticates identity and observes completion of an exam using specialized software and recordings;
b. automated proctoring where the exam session is recorded and AI (artificial intelligence) analyzed;
c. browser lockdown that limits access to other applications, websites, copying, printing, screen capture and other functions; or
d. a combination of both live/automated proctoring and browser lockdown.

Course instructors will review recordings, analyses, and data obtained through online proctoring for academic integrity infractions. It is the student's responsibility to meet the technical, software, location, and identity verification requirements necessary to enable online proctoring.

Further details of these policies are available in the Academic Calendar and on the Bow Valley College website, bowvalleycollege.ca.

Learners are encouraged to keep a copy of this course outline for future reference.

**Collection of Personal Information:**

This course, including your image and voice, may be recorded and made available to you and other students taking the course section. By attending the class(es) online or in person, you consent to the collection of your personal information. If you do not wish to be recorded, please contact your instructor before starting the course/class to discuss alternative arrangements.

You may use the recordings only for educational purposes and you must not copy, share, or use the recordings for any other purpose without the instructor's express permission.

Your personal information is collected in accordance with section 33(c) of the Freedom of Information and Protection of Privacy Act (Alberta) to deliver academic programming, support learner flexibility, promote universal design for learning principles, and for purposes consistent with the course activities and outcomes. If you have any questions about the collection, disclosure, use, or protection of this information, please contact the College's Access and Privacy Officer at foip@bowvalleycollege.ca.